```
[root@localhost Desktop]# tail -4 /etc/group
slocate:x:21:
u1:x:500:
u2:x:501:
mygroup:x:502:
[root@localhost Desktop]#
```

Figure 2: /etc/group

Tip: */etc/passwd-* and */etc/shadow-* are the *backup* files of */etc/passwd* and */etc/shadow*, respectively.

You can switch between user sessions by using the *su* command as *#su - <username>*

For example:

```
#su - u1
```

Home directories of users exist under */home*. The same can be checked by listing the contents of */home directory* using command *#ls /home*. Directories of *u1* and *u2* will be displayed.

Information about existing groups is present in the */etc/group* file and has the following format:

```
<groupname>:< auth_check >:<groupid>:<secondarymemberslist>
```

Let us have a look at the */etc/group* file (Figure 2).

Here, no group is allotted as secondary to any user but soon we will see how to allocate secondary groups.

Group passwords are stored in the */etc/gshadow* file.

Now the question is: where do the rules of default configurations reside? This is important for customisation.

The following files answer this question:

1. */etc/login.defs:* Provides control over password aging, minimum-maximum allowable user IDs/group IDs, home directory creation and other settings.
2. */etc/default/useradd:* Provides control over the home directory path, shell to be provided, kit-providing directory and other settings.
3. */etc/skel:* This is the default directory acting as a start-up kit, which provides some pre-created files for new users.

Customising default configurations for a new user

It is always good to tweak the default configurations for several reasons. It provides better security since the default settings are known to almost every admin. It also creates a customised environment that suits your requirements. It even helps you to exhibit your advanced administration skills.

Let us customise the following default settings for any new user:

- Change the default home directory path.
- Change the permitted maximum *user/group* ID range.
- Change the location of the default start-up kit source.

To do this, edit *login.defs* and make the changes as shown:

```
# vim  /etc/login.defs
```

Now, edit the *useradd* file:

```
# vim  /etc/default/useradd
```

Next, create the */ghar* and */etc/startkit* directories with some files in *startkit*.

So, we have created a file named *Welcome*, which will be available in the home directory of all new users. Now, add a new user *u3* and check its home directory under */ghar* instead of */home*.

We can see that *u3* received the file named *Welcome* in the home directory automatically. Now, when we try to add more users, the effect of *Min* and *Max* user/group ids will not allow us to accomplish user/group creation. We cannot create more users since the maximum range of the UID has been reached, which can be verified in */etc/passwd*.

Other alterations are also visible in */etc/passwd* as per customisations.

Dynamically customising default configurations for a new user

We can also make changes in configurations for individual users during new user creation. The following are the options for *useradd* and *usermod* commands:

```
-u  ->  uid -d  -> home directory -g  ->  primary group
-s  ->  shell  -G   ->  Secondary Group
```

Let us add a new user called 'myuser' with the user ID as '2014', the home directory as '/myhome', the primary group as 'u1', secondary group as 'u2' and shell as 'ksh'. Any of these options can be omitted, as required.

```
# useradd  -u  2014 -d /myhome -g  u1 -G u2 -s /sbin/ksh
myuser
```

Tip: Manual UID and GID do not adhere to minimum and maximum ID rules.

Let us verify the user information in */etc/passwd*.

Also, look at the */etc/group* file to see the effect of secondary group membership of 'myuser' with 'u2'.

Tip: After customising the primary group through the command, a new group with the name of the user is not created.

Similarly, a group can also be configured with a specific ID during creation: